

Internal Role Stability in Voynichese

Morphological Foundations and Multi Transcription Controls

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Abstract

This paper studies internal role stability in the Voynich Manuscript without proposing decipherment, plaintext language, or translation. The analysis stops at the role layer. It asks whether written units in Voynichese keep stable internal roles after morphology aware grouping, multiple transcription support, controls for section or illustration code, Currier like class, hand like class, and line marker family, and sensitivity checks over tokenization assumptions. The evidence base is a multi transcription supported core of 1,709 lines and 7,107 token occurrences. The paper builds role test units from surface tokens, conservative morphology units, broad shape units, Stolfi inspired signatures, edge signatures, and boundary merge variants. It tests role stability with threshold sensitivity, ablation tests, random role baselines, held out validation, transcription tier checks, section level profiling, independent cluster induction, predictive held out validation, and tokenization robustness. The final reporting layer contains 23 globally reliable role units, 13 conditioned reliable role units, 9 environment sensitive reliable role units, and 9 tokenization robust reliable role units. The promoted reliable layer covers 1,903 of 7,107 token occurrences, or 26.8 percent of the analyzed core, and appears in 812 of 1,709 lines, or 47.5 percent of analyzed lines. A further good but not reliable enough layer covers 272 additional token occurrences. With this descriptive layer included, coverage reaches 2,175 token occurrences and 858 lines. These are internal distributional objects. They are not word meanings, parts of speech, formal grammar rules, or semantic claims.

Keywords Voynich Manuscript, Voynichese, undeciphered scripts, computational linguistics, morphology, role structure, transcription robustness, confound control, digital humanities.

1 Introduction

The Voynich Manuscript is an illustrated codex written in an undeciphered script and preserved as Beinecke MS 408 at Yale University Library [1]. Voynichese has long been known to have non random statistical structure. It also has major confounds. These include section, Currier variety, scribal hand, layout, line position, and uncertainty about what visible spaces mean [2, 8]. A role study must therefore avoid a simple error. It must not treat frequency, location, or word shape as proof of grammar.

This paper addresses that problem at the role layer. A role is not a meaning. A role is not a part of speech. A role is an internal distributional label assigned by earlier internal annotation work and then tested here for stability. The paper asks which written units behave as stable role units after the main known confounds are treated as controls.

The contribution is a maximal reliable catalog under the current data constraints. The word maximal does not mean complete. It means that the paper reports every unit that can be promoted under the stated rules, and also reports good but unpromoted units separately. A unit may be globally reliable, conditioned reliable, or environment sensitive reliable. The first class is portable across controls. The second class is reliable inside a stated manuscript environment. The third class is role stable and independently supported, but too concentrated in one environment to be called portable. This policy accepts that role behavior can be local without treating every local pattern as reliable.

The paper makes six contributions. First, it states the transcription and unit assumptions that support the analysis. Second, it uses prior Voynich word structure work as the morphology foundation for role test units. Third, it evaluates stability under sensitivity tests, ablations, random baselines, held out validation, transcription tier checks, and section profiles. Fourth, it adds independent cluster induction and held out role prediction as tests of whether the role layer can be recovered from morphology and context. Fifth, it separates reliable, good but not reliable enough, and low reliability material so that coverage can expand without weakening the main claim. Sixth, it tests whether promoted role units remain supported when visible spaces are not treated as the only possible unit boundary.

2 Layer boundary

The paper covers the layers shown in Table 1. Formal grammar and semantics are outside the claim. Formal grammar would test ordered rules, inversions, terminal behavior, mode switching, and cross domain constraints. Semantics would require external anchors from images, layout, diagrams, repeated manuscript contexts, or medieval technical traditions.

Table 1: Layer boundary for the present paper

Layer	Status in this paper
Transcription	Used as evidence assumption
Morphology	Operational foundation
Token families	Role test units
Role stability	Main claim
Formal grammar	Future work
Semantics	Future work

This boundary is necessary because role stability is a weaker claim than grammar. A unit may keep the same internal role across contexts without being a noun, verb, operator, quantifier, or semantic action. The present paper keeps those labels out of the evidentiary claim.

3 Related work

Currier identified large scale textual differences that later work treats as major structural varieties [2]. IVTFF and related transcription work provide the multi witness setting in which different machine readable versions can be compared [3]. Stolfi developed detailed models of Voynichese word structure, including prefix, midfix, suffix analysis and later word grammar work [4, 5]. Computational studies have examined statistical structure, entropy, co occurrence patterns, and topic structure [6, 7, 9]. Bown and Lindemann summarize the linguistic state of the field and emphasize that the manuscript

remains undeciphered [8]. Recent work also cautions that glyphs, tokens, and spaces should not be treated as ordinary letters, words, and spaces without argument [11].

Table 2 positions the present paper against these prior layers. The goal is not to replace word grammar, transcription work, or entropy work. The goal is to use them as constraints on a narrower role stability question.

Table 2: Relationship to prior structural work

Prior layer	Contribution used here	What this paper adds
Currier	Textual variety is a major confound	Controls role claims against variety
IVTFF	Multiple transcription witnesses matter	Uses supported tiers rather than one stream
Stolfi	Word internal form is structured	Builds role units from morphology aware lenses
Entropy and topics	Large scale structure is measurable	Tests role stability at unit level
Recent unit cautions	Tokens and spaces are uncertain	Avoids lexical and grammatical claims

4 Evidence base

The working input is a line level Voynich transcription table produced by the project migration package. It contains 5,376 line records. The primary analysis uses the S1 plus S2 core of 1,709 lines. The expanded token occurrence table contains 7,107 token occurrences across 2,597 surface token types.

Table 3 states how transcription evidence is used. The paper does not package raw third party transcription witnesses as new source material. It uses derived project tables and records the external transcription infrastructure as an evidence assumption. This avoids treating a single transcription stream as final.

Table 3: Transcription assumptions

Item	Treatment
S1 tier	Strict multi transcription core
S2 tier	Strong but partial support
Raw witnesses	Documented as external sources
Unit uncertainty	Handled through morphology lenses and tier checks

Each token occurrence retains locus, folio, line order, token position, token string, internal role, module label, section or illustration code, Currier like class, hand like class, line marker family, and transcription support tier. The combined control stratum is the four field tuple made from section or illustration code, Currier like class, hand like class, and line marker family.

5 Morphology foundation

Role analysis should not begin with uninterpreted strings alone. Voynichese word forms have strong internal constraints, and prior work has modeled those constraints at the word grammar layer [5]. The present paper therefore builds four role test lenses. These lenses are not claimed to be a complete morphology of Voynichese. They are operational ways to test whether role stability is visible at different unit granularities.

Table 4: Role test lenses

Lens	Unit tested
surface token	Exact token string
conservative morphology	Body, terminal class, and reduced body shape
broad shape	Reduced body shape and terminal class
Stolfi inspired signature	Gallows pattern, reduced remainder shape, and terminal class

The conservative morphology lens groups tokens by limited frame material, body, terminal class, and reduced body shape. The broad shape lens groups reduced body shape and terminal class. The Stolfi inspired signature lens groups gallows pattern, reduced remainder shape, and terminal class. These lenses connect role testing to known word internal structure while avoiding a claim that the exact historical morphology is solved.

6 Internal role layer

The role labels tested in this paper were produced by earlier internal annotation work. That annotation work clustered token behavior by local context, line position, recurrence, and module structure. This paper does not ask the reader to accept those labels as semantic categories. It tests whether the labels behave as reproducible internal variables.

For each unit under each lens, the analysis computes total occurrences, distinct lines, distinct token types, dominant internal role, dominant role share, normalized role entropy, weighted within stratum role purity, agreement between local stratum majority role and global dominant role, number of combined strata, number of section or illustration codes, metadata concentration values, and maximum metadata concentration risk.

A unit is called role stable across confounds under the standard threshold only when it has enough frequency, enough line support, high dominant role share, high within stratum role purity, high stratum majority agreement, adequate spread across strata, adequate spread across sections, and low maximum metadata concentration risk. For morphology based lenses, at least two surface token types are required. This prevents a family result from merely restating one surface token.

7 Evaluation design

The evaluation is designed around skeptical alternatives. A stable unit could be a frequency artifact. It could be a line position artifact. It could be section, Currier, hand, morphology, or threshold choice. Threshold sensitivity means that the analysis is rerun under strict, standard, and loose promotion cutoffs. If a unit appears only at one cutoff, it is treated with more caution.

The paper uses eight tests. First, threshold sensitivity compares strict, standard, and loose criteria. Second, ablation tests show how many candidates survive when controls are added. Third, random role shuffle baselines test whether stable units appear after role labels are destroyed globally or within control strata. Fourth, held out validation defines units on development data and tests whether the same dominant role appears in held out folios or lines. Fifth, transcription tier robustness checks whether the full supported units are seen and role consistent in the S1 and S2 evidence tiers. Sixth, independent cluster induction groups units from morphology and context features without using role labels as input features. Seventh, predictive validation asks how well held out role labels can be recovered from morphology and context features without using manuscript section, Currier

class, hand class, or line marker family as input features. Eighth, tokenization robustness tests whether the same role support appears under surface token, morphology, shape, edge, and boundary merge variants.

These tests do not prove grammar or meaning. They ask whether the role layer is strong enough to be a reproducible object for later grammar work.

8 Core role stability results

Table 5 gives the standard threshold counts. The surface token lens yields 16 role stable units. The conservative morphology lens yields 1. The broad shape lens yields 4. The Stolfi inspired signature lens yields 8. The morphology based counts are smaller because each supported family must contain more than one surface type and must survive the same portability requirements.

Table 5: Standard role stability counts by lens

Lens	Role stable	Local	Contextual	Low power
broad	4	11	62	1029
cons morph	1	5	61	1591
stolfi sig	8	11	86	1396
surface	16	37	64	2480

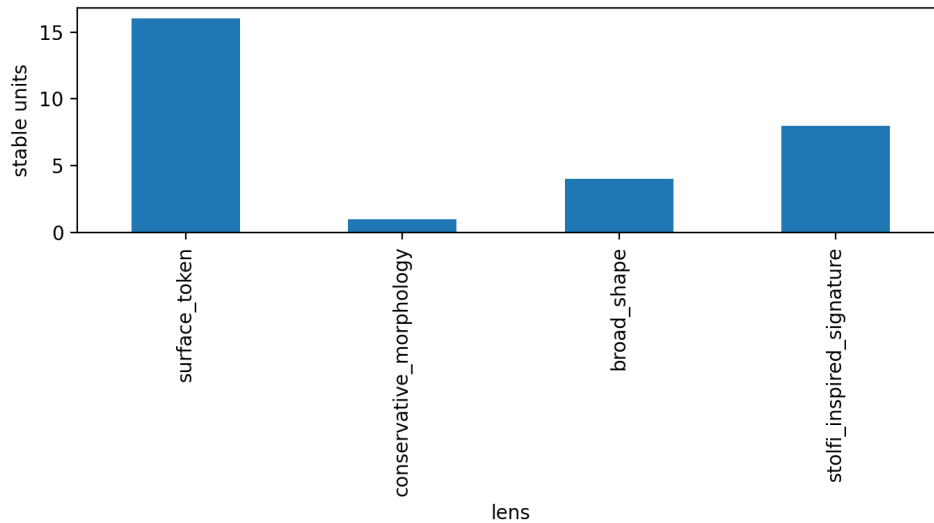


Figure 1: Stable role units by lens

Table 6 shows that the result is threshold sensitive, but not threshold arbitrary. The strict threshold retains a smaller core. The loose threshold adds a wider candidate set. The all threshold core is small, which is expected for an undeciphered manuscript with uncertain units.

Table 6: Threshold sensitivity

Lens	Strict	Standard	Loose	All three
broad	1	4	13	1
cons morph	0	1	4	0
stolfi sig	1	8	15	1
surface	7	16	32	7

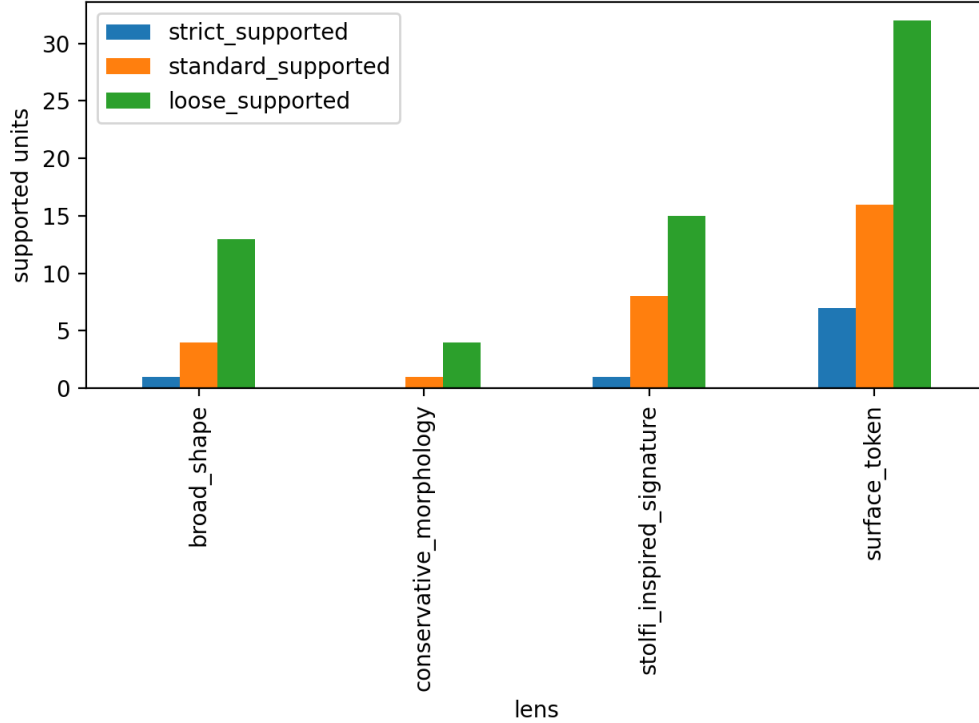


Figure 2: Strict, standard, and loose threshold comparison

Table 7 is the main control table. It shows that many candidates satisfy a role only criterion, but fewer survive line marker and full combined controls. This pattern supports the narrow claim that the final stable set is not only a frequency or role purity list.

Table 7: Ablation counts under the standard threshold

Lens	Role only	Currier	Line marker	Full
broad	7	6	4	4
cons morph	2	2	1	1
stolfi sig	10	9	8	8
surface	45	37	18	16

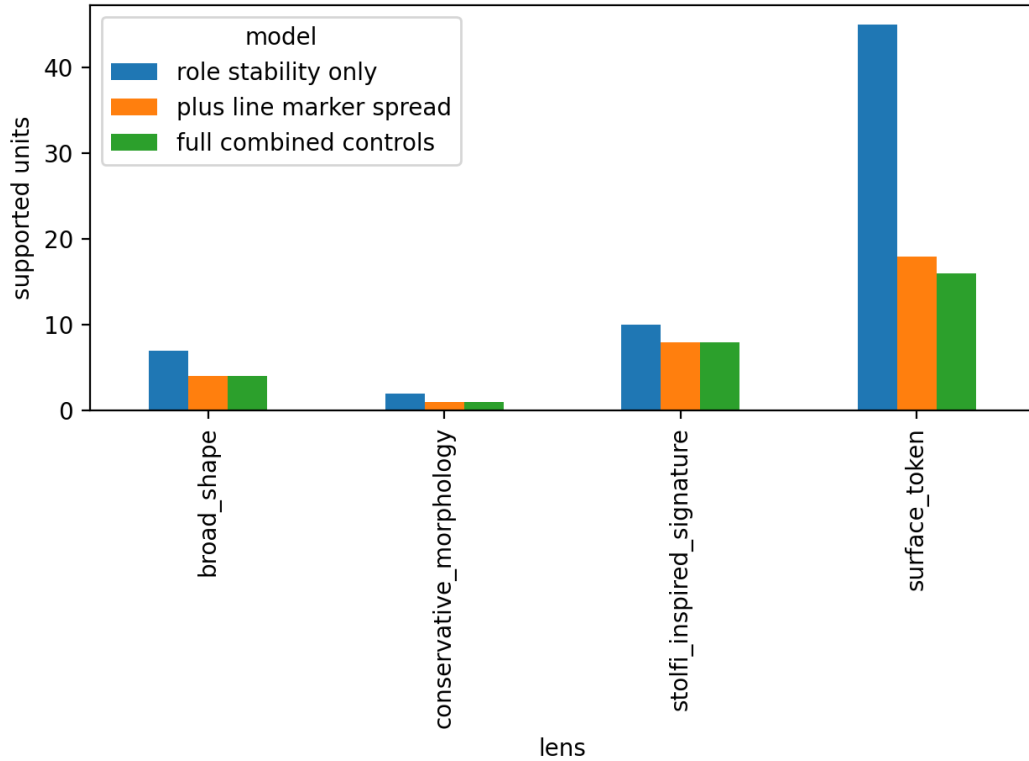


Figure 3: Effect of selected controls on supported units

9 Baseline and held out validation

Table 8 reports random role baselines. The global role shuffle baseline recovers no stable units. The within stratum shuffle baseline recovers almost none, with a maximum of one broad shape unit across the runs. This indicates that the observed stable role units depend on the actual alignment between units and roles.

Table 8: Random role shuffle baselines

Lens	Global shuffle max	Within stratum max
broad	0	1
cons morph	0	0
stolfi sig	0	0
surface	0	0

Table 9 reports held out validation. Units supported on development data are usually seen in held out data when they have enough coverage, and every seen development supported unit preserves the same dominant role. The retention column is more conservative because held out subsets are smaller and therefore must satisfy loose held out thresholds.

Table 9: Held out validation

Basis	Lens	Dev	Seen	Same role	Retained
folio	broad	1	1	1	1
folio	stolfi sig	6	6	6	3
folio	surface	10	10	10	0
line	broad	3	3	3	0
line	cons morph	1	1	1	0
line	stolfi sig	5	5	5	2
line	surface	10	10	10	1

Table 10 reports transcription tier robustness. Every full supported unit is seen in both the S1 and S2 tiers and keeps the same dominant role when seen. Standard tier support is strongest in S1. S2 is smaller and therefore retains fewer units under frequency thresholds, but it still preserves dominant role identity for the supported units that appear there.

Table 10: Transcription tier robustness

Tier	Lens	Full	Seen	Same role	Loose
S1	broad	4	4	4	4
S1	cons morph	1	1	1	1
S1	stolfi sig	8	8	8	5
S1	surface	16	16	16	14
S2	broad	4	4	4	0
S2	cons morph	1	1	1	0
S2	stolfi sig	8	8	8	4
S2	surface	16	16	16	2

10 Independent role recovery

The independent induction stage adds two tests aimed at the strongest remaining objection. The objection is that the previous results might only restate the upstream role annotation layer. The first test clusters role test units from morphology and context features without using old role labels as features. After clustering, the old role labels are used only for evaluation. Table 11 shows the number of standard role stable units whose induced cluster agrees with the audited role in a majority of four cluster resolutions.

Table 11: Independent induction agreement

Lens	Standard units	Majority agreement	All resolutions	Mean purity
broad	4	4	1	0.604
cons morph	1	1	0	0.466
stolfi sig	8	6	3	0.566
surface	16	12	8	0.733

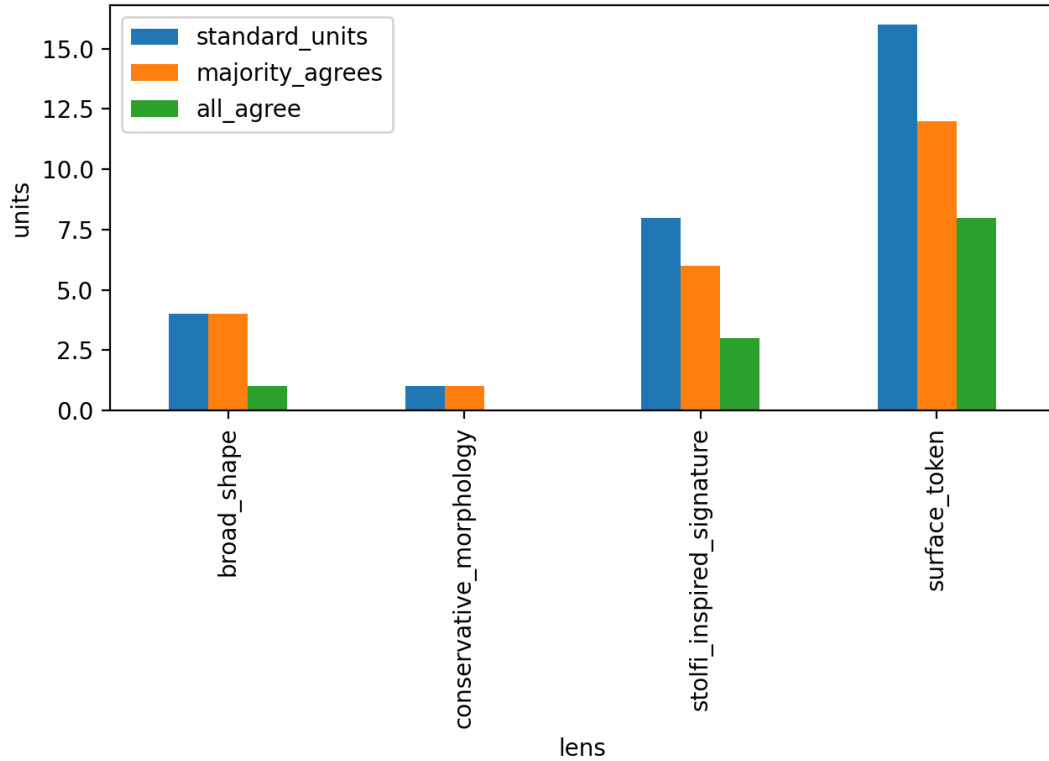


Figure 4: Agreement between induced clusters and audited roles

The second test is predictive rather than clustering based. It trains on one subset and predicts held out role labels using morphology and context features. It does not include section, Currier like class, hand like class, or line marker family as input features. Table 12 shows that morphology and context substantially outperform majority and position only baselines. This does not prove that the roles are linguistic categories. It does show that the role layer is recoverable from written form and local context rather than only from manuscript metadata.

Table 12: Held out role prediction

Basis	Model	Acc	Bal acc	Macro F1	Weighted F1
folio	majority role baseline	0.396	0.038	0.022	0.224
folio	position only	0.142	0.068	0.033	0.2
folio	morphology only	0.875	0.818	0.761	0.878
folio	morphology plus context	0.88	0.788	0.745	0.88
line	majority role baseline	0.387	0.042	0.023	0.216
line	position only	0.137	0.052	0.034	0.197
line	morphology only	0.897	0.915	0.856	0.901
line	morphology plus context	0.886	0.8	0.738	0.887

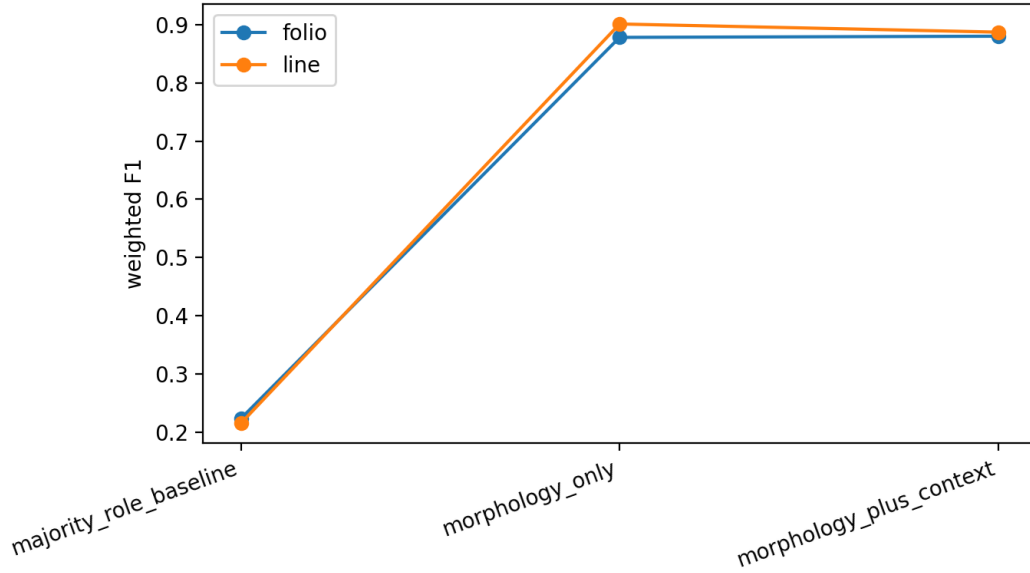


Figure 5: Held out role prediction by feature set

11 Global and conditioned reliable roles

A globally reliable unit must satisfy the standard confound controls and must also agree with independently induced clusters in at least two of the four cluster resolutions. Under this policy, the global catalog contains 23 globally reliable role units. The global catalog covers 769 unique token occurrences, or 10.8 percent of the S1 plus S2 core. It appears in 509 lines, or 29.8 percent of the core. This is high precision global coverage rather than full annotation coverage.

Table 13: Maximal reliable role catalog

Lens	Unit	Role	Occ	Lines	Types	Ind	Tokens
broad	X NONE	R06	46	46	5	3/4	s, x, g, c
broad	O dy	R00	43	43	8	3/4	chody, shody, otody, okody
broad	X y	R06	30	29	4	2/4	shy, sy, osy, olsy
broad	OG y	R00	26	24	8	4/4	choky, choty, otoky, choppy
cons morph	o dy O	R00	41	41	6	2/4	chody, shody, otody, okody
stolfi sig	NG S_EY T_y	R06	104	96	4	4/4	chey, shey, ey, qey
stolfi sig	NG S_OEL T_in	R06	93	82	9	4/4	dain, sain, ain, shain
stolfi sig	NG S_EY T_ey	R06	63	58	6	4/4	sheey, cheey, qeey, deey
stolfi sig	NG S_X T_NONE	R06	46	46	5	2/4	s, x, g, c
stolfi sig	NG S_BY T_y	R00	33	32	9	3/4	chy, olchy, dshy, dchy
stolfi sig	NG S_XY T_y	R06	30	29	4	3/4	shy, sy, osy, olsy
surface	ar	R06	58	52	1	4/4	ar
surface	dar	R06	49	47	1	4/4	dar

Lens	Unit	Role	Occ	Lines	Types	Ind	Tokens
surface	dain	R06	48	45	1	4/4	dain
surface	or	R06	48	46	1	4/4	or
surface	shey	R06	48	46	1	4/4	shey
surface	okaiin	R10	42	41	1	2/4	okaiin
surface	s	R06	39	39	1	4/4	s
surface	dal	R09	37	36	1	2/4	dal
surface	okain	R10	36	34	1	2/4	okain
surface	okal	R07	36	34	1	2/4	okal
surface	sheey	R06	29	28	1	4/4	sheey
surface	y	R06	25	25	1	4/4	y
surface	daiin	R10	78	67	1	NA	daiin

The conditioned reliability stage adds conditioned reliable units. These are not globally portable claims. They are stable only inside a stated manuscript environment, mostly Currier class, and are reported separately so that reliable local structure is not missed.

Table 14: Conditioned reliable units

Scope	Lens	Unit	Role	Occ	New	Lines	Examples
Currier B	surface	shedy	R06	95	95	83	shedy
Currier B	surface	chedy	R02	94	94	82	chedy
Currier B	surface	qokedy	H12	89	89	73	qokedy
Currier A	surface	daiin	R10	81	81	69	daiin
Currier B	surface	qokeey	H08	66	66	56	qokeey
Currier B	broad_shape	LB edy	R00	44	44	42	lchedy, lshedy, qolchedy, rchedy, qorchedy
Currier B	surface	qokaiin	H08	37	37	36	qokaiin
Currier B	surface	okedy	R00	33	33	29	okedy
Currier B	surface	okeedy	R00	32	32	30	okeedy
Currier B	surface	otedy	R01	30	30	28	otedy
Currier B	surface	saiin	R06	28	28	28	saiin
Currier B	surface	qokar	R03	26	26	24	qokar
Currier B	surface	chdy	R02	25	25	24	chdy

12 Environment sensitive expansion

The environment sensitive expansion responds to a stricter version of the role question. Global portability is sufficient evidence, but it is not necessary evidence. Some role units may be real while being concentrated in one textual environment. It therefore adds environment sensitive reliable units when three conditions hold. The unit must have high role purity, enough occurrence and line support, and independent induction support from morphology and context features. Metadata concentration is allowed, but only because the unit is reported as environment sensitive rather than globally portable.

Table 15: Environment sensitive reliable units

Scope	Lens	Unit	Role	Occ	Lines	Votes	New	Top tokens
localized global	surface token	ol	R06	73	66	5	73	ol
localized global	surface token	aiin	R06	57	51	5	57	aiin
localized global	surface token	chckhy	R00	41	39	4	41	chckhy
localized global	surface token	al	R06	31	28	5	31	al
localized global	surface token	oky	R00	29	29	5	29	oky
localized global	surface token	sheedy	R06	23	22	5	23	sheedy
localized global	broad shape	OXGX y	R00	23	23	4	23	chocthy, chockhy, shocthy, chocfhy, shockhy
localized global	surface token	sho	R06	22	21	5	22	sho
localized global	surface token	saiin	R06	39	38	5	11	saiin

The added environment sensitive layer contributes 310 new token occurrences and 257 lines before overlap. Together, the global reliable, conditioned reliable, and environment sensitive reliable tiers cover 1,759 token occurrences, or 24.8 percent of the core. They touch 759 lines, or 44.4 percent of the core.

13 Tokenization robustness and final coverage

Visible space tokenization is a defensible starting point, but it is not treated as final. The tokenization robustness stage tests whether candidate role units remain supported when the unit boundary is varied. The tested variants include surface tokens, conservative morphology units, broad shape units, Stolfi inspired signatures, frame and terminal units, token edge signatures, and left or right boundary merge signatures. The boundary merge checks ask whether a role signal depends on treating the visible space on either side of the unit as fixed.

A tokenization robust unit is promoted only if it keeps high role purity, has independent induction support, and has tokenization support from at least two unitization variants. This rule lets coverage rise, but it blocks candidates that look good only under one segmentation assumption.

Table 16: Tokenization robustness summary

Lens	Candidates	Induction	Tokenization	Both
broad shape	731	154	52	29
conservative morphology	550	134	15	9
stolfi inspired signature	703	229	87	68
surface token	533	163	209	152

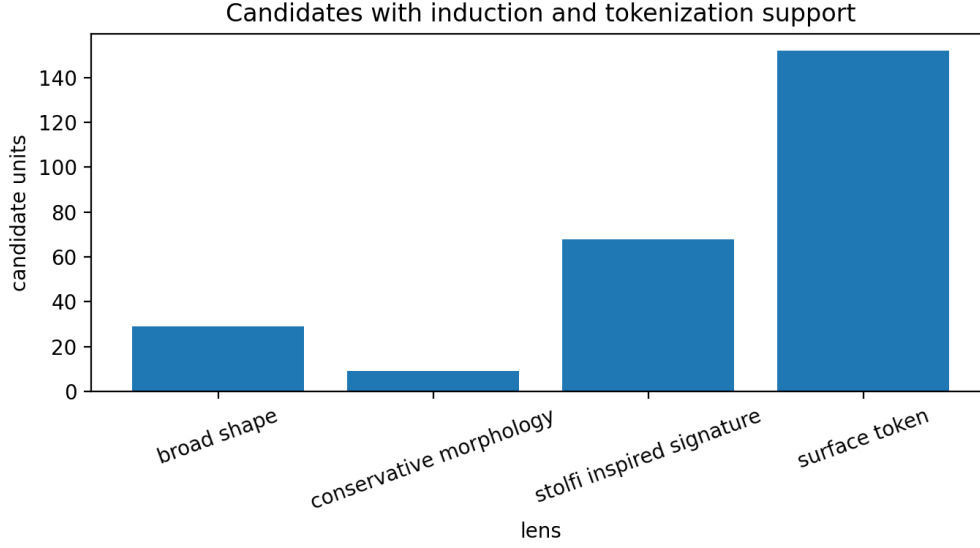


Figure 6: Candidate units with both induction and tokenization support

Table 17: Tokenization robust reliable units

Scope	Lens	Unit	Role	Occ	Lines	Ind	Tokn	New	Top tokens
global	broad shape	O NONE	R00	18	18	5	7	18	o, a, oa
hand 2	stolli inspired signature	NG S_OL T_r	R06	82	64	5	4	24	dar, or, ar, sar, sor
global	surface token	sol	R06	15	15	5	3	15	sol
section Z	broad shape	OL y	R00	15	15	5	3	15	otaly, okaly, okary, ary, okoly
hand 4	broad shape	EO dy	R00	15	15	5	3	15	okeody, cheody, sheody, okeeeody, oeeody
section H	conservative morphology	ckh y XGX	R00	20	20	5	2	10	chckhy, ckhy, ockhy, qockhy, shckhy
global	surface token	l	R06	15	15	2	3	15	l
global	stolli inspired signature	G_k S_LGEY T_ee	R00	17	15	2	2	17	lkeey, lkeey, rkeey, qolkeey
global	surface token	oly	R06	15	15	2	2	15	oly

The tokenization robust layer adds 144 new promoted token occurrences and 133 lines before overlap. The full promoted layer now covers 1,903 token occurrences, or 26.8 percent of the core. It touches 812 lines, or 47.5 percent of the core. The good but not reliable enough tier adds 272 further occurrences for descriptive coverage only. With that tier included, the descriptive layer covers 2,175 token occurrences and touches 858 lines.

Table 18: Coverage by reliability tier

Tier	Tok	Tok %	Lines	Line %	Cum tok	Cum tok %	Cum lines	Cum line %
Global reliable	769	10.82	509	29.78	769	10.82	509	29.78
Conditioned reliable	680	9.57	382	22.35	1449	20.39	693	40.55
Environment sensitive	310	4.36	257	15.04	1759	24.75	759	44.41
Tokenization robust	144	2.03	133	7.78	1903	26.78	812	47.51
Good not reliable	272	3.83	219	12.81	2175	30.60	858	50.20
Low or unassigned	4932	69.40	1584	92.69	7107	100.00	1709	100.00

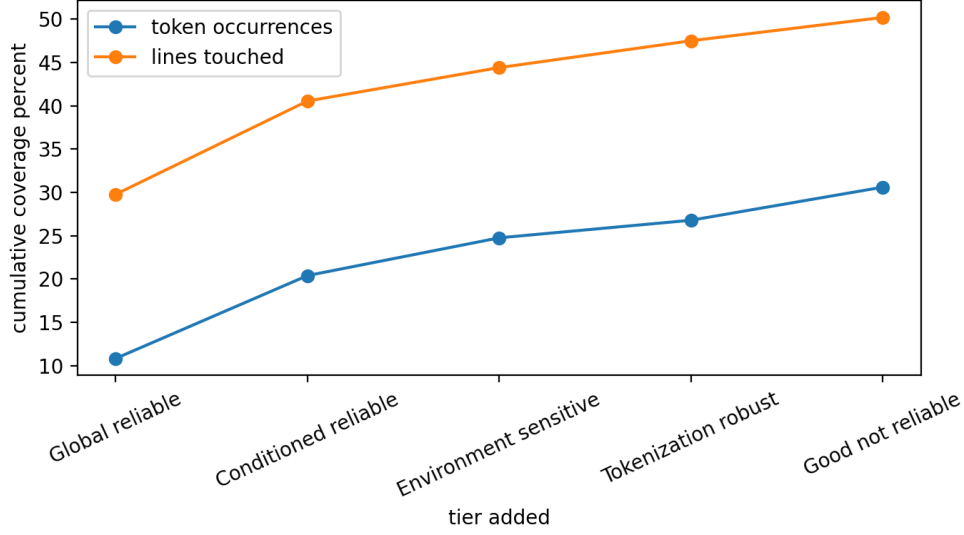


Figure 7: Cumulative coverage after adding each reliability tier

The next tier is good but not reliable enough. These units are retained so that plausible role behavior is not lost, but they are not used as promoted evidence. The common reasons for non promotion are low induction support, low tokenization support, lower role purity, or too little evidence.

Table 19: Largest good but not reliable enough units

Scope	Lens	Unit	Role	Occ	Lines	Ind	Token	New	Reason
global	surface token	daiin	R10	134	116	1	2	53	low induction
line_marker paragraph_continuation	surface token	daiin	R10	126	108	1	2	50	low induction
global	broad shape	EXGX y	R00	25	25	0	2	25	low induction
currier B	stolfi inspired signature	NG S_OL T_r	R06	143	113	5	2	23	low purity
hand 2	surface token	daiin	R10	23	19	1	2	23	low induction
hand 2	surface token	qokal	H08	23	21	1	2	23	low induction
line_marker paragraph_continuation	broad shape	EXGX y	R00	23	23	0	2	23	low induction
currier_hand B 2	surface token	daiin	R10	23	19	1	2	23	low induction
currier_hand B 2	surface token	qokal	H08	23	21	1	2	23	low induction
currier B	broad shape	EXGX y	R00	21	21	0	2	21	low induction
currier_marker B paragraph_continuation	broad shape	EXGX y	R00	21	21	0	2	21	low induction
currier_marker B paragraph_continuation	stolfi inspired signature	NG S_OL T_r	R06	137	107	5	2	20	low purity

Table 20: Role coverage across reliability tiers

Tier	Role	Tok	Tok %	Lines	Top tokens
Global reliable	R00	102	1.44	95	chy, chody, choky, shody, okody
Global reliable	R06	516	7.26	381	ar, chey, dar, shey, or
Global reliable	R07	36	0.51	34	okal
Global reliable	R09	37	0.52	36	dal
Global reliable	R10	78	1.10	72	okaiin, okain
Conditioned reliable	H08	103	1.45	88	qokeey, qokaiin
Conditioned reliable	H12	89	1.25	73	qokedy
Conditioned reliable	R00	109	1.53	97	okedy, okeedy, lchedy, lshedy, qolchedy
Conditioned reliable	R01	30	0.42	28	otedy
Conditioned reliable	R02	119	1.67	103	chedy, chdy
Conditioned reliable	R03	26	0.37	24	qokar
Conditioned reliable	R06	123	1.73	104	shedy, saiin
Conditioned reliable	R10	81	1.14	69	daiin
Environment sensitive	R00	93	1.31	87	chckhy, oky, chocthy, chockhy, shocthy
Environment sensitive	R06	217	3.05	178	ol, aiin, al, sheedy, sho
Tokenization robust	R00	75	1.06	73	o, lkeey, okeody, ckhy, lkeey
Tokenization robust	R06	69	0.97	61	sol, l, oly, sar, sor
Good not reliable	H08	23	0.32	21	qokal
Good not reliable	R00	91	1.28	84	lor, olkeedy, checkhy, chos, okary
Good not reliable	R06	105	1.48	99	dol, dam, sheol, cheedy, char
Good not reliable	R10	53	0.75	47	daiin

14 Section level roles and role switching

The paper also reports section level role profiles. The purpose is to avoid an over uniform claim. A unit may be portable while the manuscript still has section level differences. Table 21 and Figure 8 show that dominant role mixtures differ by section code. This supports the use of section as a control and also provides a descriptive layer for future grammar work.

Table 21: Section level role profiles

Section	Occ	Lines	Top role	Share	Second	Share
H	1893	340	R06	0.353	R00	0.339
S	1872	207	R00	0.407	R06	0.27
B	1512	271	R00	0.335	R06	0.295
T	467	96	R06	0.353	R00	0.336
P	428	216	R00	0.533	R06	0.236
C	403	197	R00	0.514	R06	0.323
A	295	161	R00	0.59	R06	0.315
Z	237	221	R00	0.789	R06	0.127

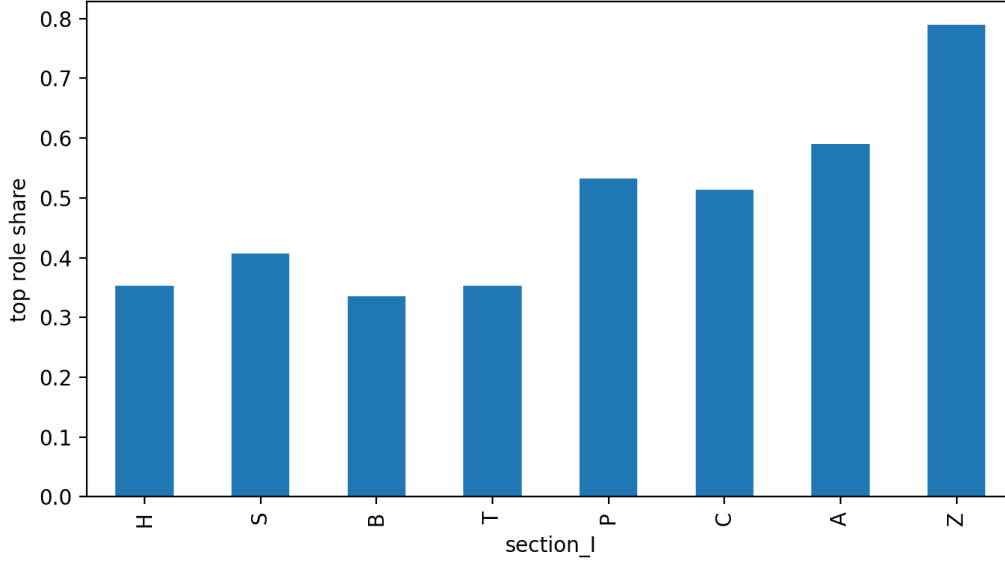


Figure 8: Top role shares by section code

A role switching check tests whether the same unit receives different reliable roles in different local scopes. The method allows such cases. Under the current promotion rules, no high confidence role switching case is found. This is a negative result rather than a rule of the system.

Table 22: Role switching audit

Test	Result
High confidence role switching	No promoted cases found
Scoped role method	Allows role differences when evidence supports them

15 Future hypotheses

The stable role units may later support stronger hypotheses. Some roles may turn out to be operator like, terminal like, quantifier like, or procedure like. The present paper does not assert those interpretations. Table 23 states them as future tests whose validity is independent of the present result.

Table 23: Future hypotheses outside the current claim

Hypothesis	Required future test
Operator like roles	Show ordered transitions and action like distribution
Terminal like roles	Show termination beyond line position effects
Quantifier like roles	Show count or measure behavior from external anchors
Procedure like structure	Show formal sequence rules across held out pages

This separation matters. If every future hypothesis in Table 23 fails, the present result can still stand. It would still show that some internal role units are stable under the stated controls. If some future hypotheses succeed, the role layer can become the foundation for a separate formal grammar paper.

16 Limitations

The main limitation is that the role labels come from an upstream internal annotation layer. Independent clustering and held out prediction reduce this dependence, but they do not fully replace the upstream layer with a new raw manuscript role induction system.

A second limitation is that the morphology layer is operational. It is based on known word structure ideas, but it is not a complete historical morphology. A third limitation is that conditioned reliable units and environment sensitive reliable units are not globally portable and should not be cited as global role evidence. A fourth limitation is that transcription tier robustness is not the same as a full leave one witness out test, because the available project table records support tiers rather than all individual witness strings. A fifth limitation is that held out tests are smaller than the full corpus and therefore conservative frequency thresholds can remove otherwise role consistent units.

These limitations define the claim. The paper establishes a role stability layer that can be independently rerun and challenged. It does not establish formal grammar or semantics.

17 Conclusion

This paper provides a morphology aware and multi transcription controlled account of internal role stability in Voynichese. The strongest result is a reliable catalog of role units that survives combined controls, threshold checks, ablations, random role baselines, held out validation, transcription tier checks, section profiling, independent cluster induction, held out predictive validation, and tokenization robustness checks. The tokenization robustness stage expands coverage by promoting only the additional units that survive alternate unit boundary checks.

The result is intentionally limited. It is not a decipherment. It does not identify a language, a lexicon, parts of speech, phonetic values, formal grammar, or semantics. Its contribution is the reproducible role layer that must come before those stronger claims.

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